



Motor intent recognition of multi-feature fusion EEG signals by UMAP algorithm

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Abstract

The key to the analysis of electroencephalogram (EEG) signals lies in the extraction of effective features from the raw EEG signals, which can then be utilized to augment the classification accuracy of motor imagery (MI) applications in brain-computer interface (BCI). It can be argued that the utilization of features from multiple domains can be a more effective approach to feature extraction for MI pattern classification, as it can provide a more comprehensive set of information that the traditional single feature extraction method may not be able to capture. In this paper, a multi-feature fusion algorithm based on uniform manifold approximate and projection (UMAP) is proposed for motor imagery EEG signals. The brain functional network and common spatial pattern (CSP) are initially extracted as features. Subsequently, UMAP is utilized to fuse the extracted multi-domain features to generate low-dimensional features with improved discriminative capability. Finally, the k-nearest neighbor (KNN) classifier is applied in a lower dimensional space. The proposed method is evaluated using left–right hand EEG signals, and achieved the average accuracy of over 92%. The results indicate that, compared with single-domain-based feature extraction methods, multi-feature fusion EEG signal classification based on the UMAP algorithm yields superior classification and visualization performance.

Keywords Motor imagery · Manifold learning · Feature fusion · Dimensionality reduction

1 Introduction

Motor imagery (MI) is a key experimental paradigm in the non-invasive BCI system, which can convert a person's mental image of a movement into control signals [1]. These

control signals can then be applied in various fields, such as brain-controlled robots, brain-controlled exoskeletons, medical diagnosis and rehabilitation treatment [2–5]. The effective extraction of distinctive features for the recognition of motor imagery tasks from complex MI EEG signals is paramount for the efficacy of BCI systems. When individuals imagine physical movements, the associated brain regions are activated, which is manifested by the decreased or increased power spectral density of MI-EEG signals, referred to as event-related desynchronization (ERD) and event-related synchronization (ERS) respectively [6]. The evidence available indicates that there is a considerable amount of similarity between actual and imagined movement on the shared functional circuit [7].

Currently, some researchers have conducted pertinent studies on MI-BCI systems and achieved remarkable outcomes. A conventional algorithm for recognizing EEG signals is composed of two main components: feature extraction and feature classification. Popular feature extraction techniques for MI EEG signals include the adaptive auto regressive (AAR) model [8], the wavelet transform (WT) [9], the empirical mode decomposition (EMD) [10], and the common spatial pattern (CSP) [11]. Despite the fact that the majority of research has

Highlights

- (1) Aiming at the limitation of single feature extraction, an analysis method based on the combination of brain functional network and CSP is proposed.
- (2) Through feature fusion and dimension reduction, low-dimensional information features with strong discriminative ability are obtained.
- (3) Compared with the traditional method, the classification performance of motor imagery EEG signals is improved.

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only used single domain feature extraction, which only describes a limited amount of data and results in unsatisfactory outcomes, combining or fusing these features can be a more effective feature extraction strategy. This is because the features from different domains can provide complementary information for classification and may have different sensitivities to different MI patterns. Moein Radman et al. used a constant-q filter to extract spatial, spectral and time-domain characteristics of EEG signals. Relief F, mRMR and Fisher techniques were then combined using the Dempster-Shafer evidence method [12]. To create a classification model for stroke EEG recordings, Li et al. developed a multi-feature fusion technique incorporating wavelet packet energy, fuzzy entropy, and hierarchical theory [13]. Li et al. proposed a multi-dimensional MI-EEG imaging method based on time–frequency analysis and the Clough-Tocher (CT) interpolation algorithm, and then designed a multilevel and multiscale feature fusion convolutional neural network for WTMI [14].

Multi-domain features have often been extracted in order to fully use more information from many domains and enhance classification performance. Nevertheless, these features usually possess high dimensionality, which can necessitate the use of a more complex model and may even diminish the accuracy of the classifier. To address this issue, uniform manifold approximate and projection (UMAP) is used to achieve information fusion and dimensionality reduction in this study. The common spatial patterns algorithm (CSP) method can extract the spatial information from EEG signals and result in a significant improvement in the classification of two-class EEG signals. The CSP algorithm is used to maximize the projection scattering difference between two classes of EEG signals by constructing an optimal spatial filter. Studies have demonstrated that the brain is composed of multiple brain regions with distinct functions, and motor imagery requires the cooperation of multiple brain regions to complete the imagined task [15]. In order to study brain functional networks, some researchers have employed the phase synchronization (PS) method. Compared to the amplitude synchronization method, PS is able to disregard the influence of EEG instantaneous amplitude and is only sensitive to phase, making it more suitable for the analysis of nonlinear and nonstationary MI-EEG signals [16]. Phase-locked value (PLV) is a well-established phase synchronization technique that has been widely used in the study of motor imagery-based functional brain networks. Therefore, it is necessary to fuse them to improve the classification ability of MI patterns. In this study, a multi-feature fusion algorithm based on nonlinear dimensionality reduction, namely UMAP [17], is utilized to extract more informative features for MI pattern classification. In the first step, spatial domain and functional brain network features are extracted from raw EEG signals to serve as initial features. Subsequently, UMAP is utilized to reduce

the dimensionality of the multi-domain features, thereby improving their discriminative power. Finally, a KNN classifier is employed to achieve pattern classification, and the performance of EEG signal recognition and feature data visualization is assessed.

The rest of the paper is organized as follows: Section 2 introduces the EEG data studied, the feature extraction techniques used and the improved methods in this paper. Section 3 presents the experimental findings and contrasts the results of the recognition before and after feature fusion. Section 4 compares the proposed method with the existing methods. Finally, the conclusions of this paper are derived in Section 5.

2 Materials and methods

2.1 Dataset

We recruited ten subjects (6 males and 4 females), all of whom have normal vision and are right-handed. Before the experiment, all subjects were informed about the purpose and the procedure of the experiment. In this study, the bowl-ball experiment was used as an EEG-evoked paradigm [18], with participants sitting approximately 60 cm away from a screen displaying a virtual bowl-ball system. Participants were required to hold the ball in the virtual bowl and control its movement from the starting point to the end point by pressing “←” and “→” on the computer keyboard. They had to ensure that the virtual ball did not overflow the edge of the bowl, or else the task would end immediately.

Subjects collected experimental data through an EPOC + EEG helmet, which is a 14-channel EEG signal acquisition device (AF3, F7, F3, FC5, T7, P7, O1, O2, P8, T8, FC6, F4, F8, AF4) with a sampling frequency of 128 Hz. The electrode position conforms to the international 10–20 electrode system. During the experiment, the computer recorded the subjects’ EEG signals and their working states, including left execution, right execution, and resting state. The data was divided through the left and right execution, and the collected data was then combined into a 4-dimensional EEG signal (number of channels × number of samples × number of trials × number of classes).

2.2 Feature extraction

In this study, the CSP algorithm and PLV-based brain network topology properties are used to extract the data features of motor imagery EEG signals. These features are then fused and reduced by the optimized uniform manifold approximation and projection algorithm, and then classified by using the KNN classifier. Finally, the classification performance and visualization performance of the feature data are evaluated

and analyzed. The CSP algorithm is capable of producing eigenvectors with enhanced discrimination by maximizing the variance differences between different signals contained in the same projection matrix. Furthermore, PLV can be used to measure the degree of synchronization between two signals that are phase-locked in a specific frequency band and over a given time period [19]. As illustrated in Fig. 1, the specific experimental framework is presented.

2.2.1 Spatial domain feature extraction

CSP is an effective algorithm for the analysis of dichotomous EEG signals [20], based on multi-channel spatial filtering technique. It utilizes the diagonalization of the covariance matrix to identify a set of optimal spatial filters for projection, maximizing the variance of one class while minimizing the variance of the other class. When the EEG signal is projected by the filter matrix, it can achieve a higher correlation with one class and a lower correlation with the other, thus achieving the purpose of distinguishing the two classes. The mathematical expression of CSP function is as follows:

$$J(\omega) = \frac{\omega^T C_1 \omega}{\omega^T C_2 \omega} \quad (1)$$

where ω is a spatial filter matrix, ω^T represents the transpose of the matrix ω , and C_1 and C_2 mean covariance matrices, respectively.

The matrix $Z_{N \times T}$ is projected by the filter matrix ω to obtain the feature matrix.

$$Z_{N \times T} = W_{N \times N} \times X_{N \times T} \quad (2)$$

where N is the number of channels of the corresponding EEG signal and T is the number of samples in a signal trial.

The feature matrix Fp is extracted from the matrix Z , where the dimension of the matrix Fp is less than or equal to the number of channels of the original EEG signal, where each row could project the data vector of one time point to one entry of the corresponding feature vector.

$$Fp = \log \frac{\text{var}(Z_i)}{\sum_{p=1}^{2m} \text{var}(Z_p)}, p = 1 \cdots 2m \quad (3)$$

where Fp is the normalized feature matrix of the p^{th} sample. The first m rows and the last m rows of the data are selected as the feature matrix ($m = 1$ to 7). The dimensionality of the feature matrix is $2m$.

2.2.2 Brain functional network construction

EEG signals are recorded as time-series signals of brain activity, and studies show that such time-series signals recorded at different brain regions reflect the brain activity synergy of their corresponding brain regions. Such EEG

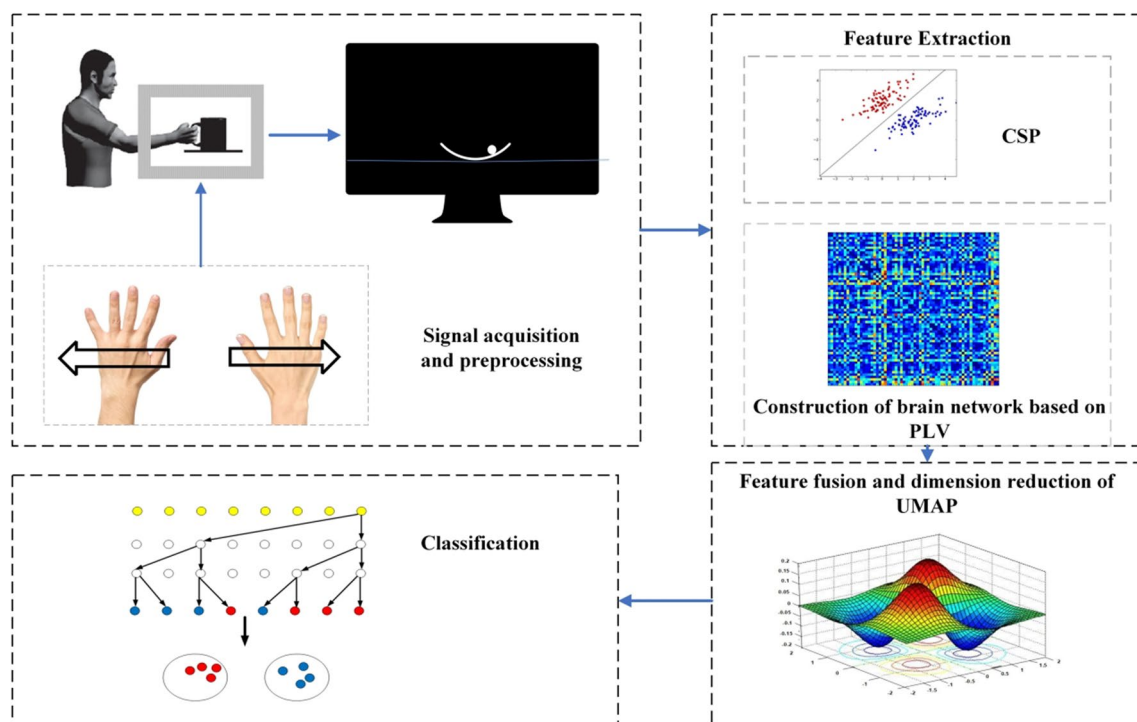


Fig. 1 The framework of proposed method

time series signals acquired from different locations in the brain form a brain network [21, 22], and by extracting various measures from this network, it is possible to identify differences in the brain regions activated by different individuals in response to various cognitive tasks.

The EEG electrode is commonly used as a node in graph theory-based research on brain functional networks, and the connections between each node act as the edges. Synchronization assessment based on phase information is more effective for information recognition, as studies have shown that EEG signals have a phase synchronization relationship in each brain area, but no amplitude synchronization relationship [23]. Therefore, PLV is selected to construct the brain network in this paper. If each electrode is considered as a node, then the node i and node j in the motor imagery EEG can be defined as the phase locking value between the n th intrinsic mode function between channel i and channel j in the EEG signal. The calculation process is shown in Eq. (4):

$$PLV_{ij}(t) = \frac{1}{S} \left| \sum_{s=1}^S \exp(i\Delta\varphi_{ij}(t)) \right| \quad (4)$$

where $\Delta\varphi_{ij}(t) = \varphi_i(t) - \varphi_j(t)$ represents the instantaneous phase difference between channel i and channel j in t^{th} trial, S is the total number of samples in the trial. The computed PLV is a number between 0 and 1 that represents how well the two channels are synchronized. The degree of synchronization between the two nodes increases as the PLV value increases.

Node degree indicates the number of adjacent nodes in the brain functional network, reflecting the connectivity between nodes, as shown in Eq. (5):

$$K_i = \sum_{j=1}^N a_{ij} \quad (5)$$

where N is the total number of network nodes, a_{ij} represents the connectivity of channel nodes, and the size of each parameter a_{ij} determines how close the relationship between node i and node j .

The clustering coefficient shows the possibility that all adjacent nodes of a node in the network are adjacent nodes to each other. According to the Eq. (6), the clustering coefficient C_i of node i is defined as the ratio of the number of connected edges to the number of most possible connected edges.

$$C_i = \frac{2E_i}{k_i(k_i - 1)} \quad (6)$$

where E_i reflects the actual number of connected edges between node i and adjacent nodes, k_i is the number of all adjacent nodes of a node, and $k_i(k_i - 1)/2$ represents the

maximum number of possible connected edges between node i and its adjacent nodes.

The betweenness centrality is defined as the number of shortest paths going through a node or an edge. The higher the betweenness centrality of a node, the greater the flow of information carried by the node, and the more significant the impact on the function of the functional brain network, as shown in Eq. (7):

$$B(u) = \sum_{s \neq u \neq r} \frac{\delta_{sr}(u)}{\delta_{sr}} \quad (7)$$

where u is the node, and $\delta_{sr}(u)$ denotes the number of all shortest paths between node u from node s to node r , $\delta_{sr}(u)$ is the length of the shortest path from node s to node r .

2.3 Feature fusion and dimensionality reduction

The UMAP algorithm is a nonlinear dimensionality reduction method based on Riemannian geometry and algebraic topology theory. It is used to learn the manifold structure in high-dimensional space and to find a low-dimensional representation of the data. As illustrated in Fig. 2, two points that are close to each other when measured in a straight line may have to be circled twice along the curve when measured in the manifold space [24]. UMAP utilizes the characteristics of the high-dimensional manifold structure to determine the positional relationship between points and construct a high-dimensional data distribution structure. After projecting the data into low-dimensional space, the distance relationship is reconstructed and the eigenvalues are reconstructed based

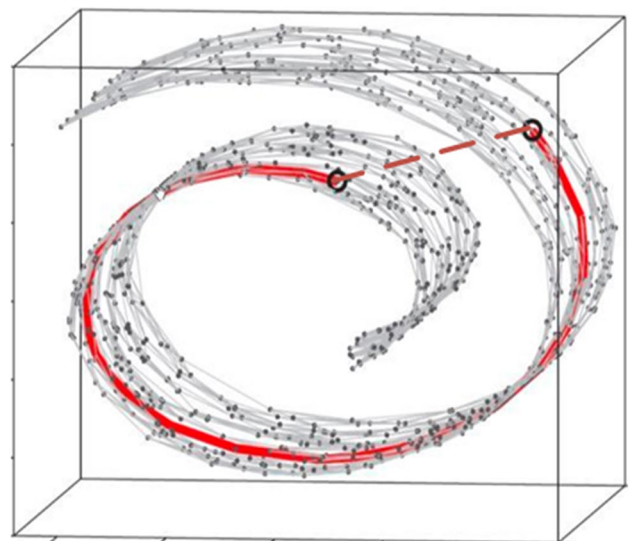


Fig. 2 Schematic diagram of manifold distance in multi-dimensional space

on the relative relationships between points in the high-dimensional space [25].

UMAP is based on three main assumptions: (1) the data are uniformly distributed on the Riemannian manifold, (2) the Riemannian metric is a local constant, and (3) the manifold is locally connected [26]. These assumptions imply that the manifold can be represented by the fuzzy topology of high-dimensional data points. By exploring the fuzzy topology of the low-dimensional projection of the data, the embedded manifold can be discovered.

As shown in Eq. (8), the experimental data of ten subjects were extracted respectively, the feature vector U_i is constructed by fusing the common spatial pattern EEG signal-based feature vector F_i and the node degree feature K_i , the clustering coefficient feature C_i and the betweenness centrality feature B_i of the functional brain network based on the phase-locked values in order.

$$U_i = [F_i K_i C_i B_i] \quad (8)$$

UMAP utilizes high-dimensional data as its input and creates a topological representation which then produces a low-dimensional representation of the input data with the least amount of structural difference by minimizing the cross-entropy between them. The cross-entropy CE of fuzzy sets (H, μ) and (H, ν) is calculated as the difference between the two sets, which is used to measure the similarity between them.

$$CE((H, \mu), (H, \nu)) \triangleq \sum_{h \in H} \left(\mu(h) \log \left(\frac{\mu(h)}{\nu(h)} \right) + (1 - \mu(h)) \log \left(\frac{1 - \mu(h)}{1 - \nu(h)} \right) \right) \quad (9)$$

3 Results

3.1 Feature extraction based on CSP

The classification accuracy of the CSP-selected feature dimension is evaluated by randomly selecting a subject and plotting the ROC curves of different CSP-projected dimensionality (m) Fig. 3. The bulge shape and proximity to the upper left corner of the figure indicate that this index can provide more evaluation value. The area under the ROC curve (AUC) is an important statistic, and the larger the AUC, the better the performance.

As shown in Fig. 3, the projection dimension ranges from 1 to 7. As the dimension (m) increases, the separability of EEG signals increases initially before reaching a peak at $m = 5$, where the highest AUC value is achieved and the separability is at its optimum. Beyond this point, the separability begins to decrease. By projecting the EEG data of other subjects, the AUC was found to reach its maximum value when the projection dimension m was set to 5. Consequently, this study selected $m = 5$ as the optimal parameter of CSP, and then used it to generate a 10-dimensional feature vector F_i from the motor imagery EEG data. As shown in Fig. 4, the CSP algorithm is employed to map the feature data into the two-dimensional space and select the most suitable spatial projection. Unfortunately, the classification and visualization of the feature data in two-dimensional space are not satisfactory.

Fig. 3 Area under curves (AUC) values of different CSP-projected dimensionality (m)

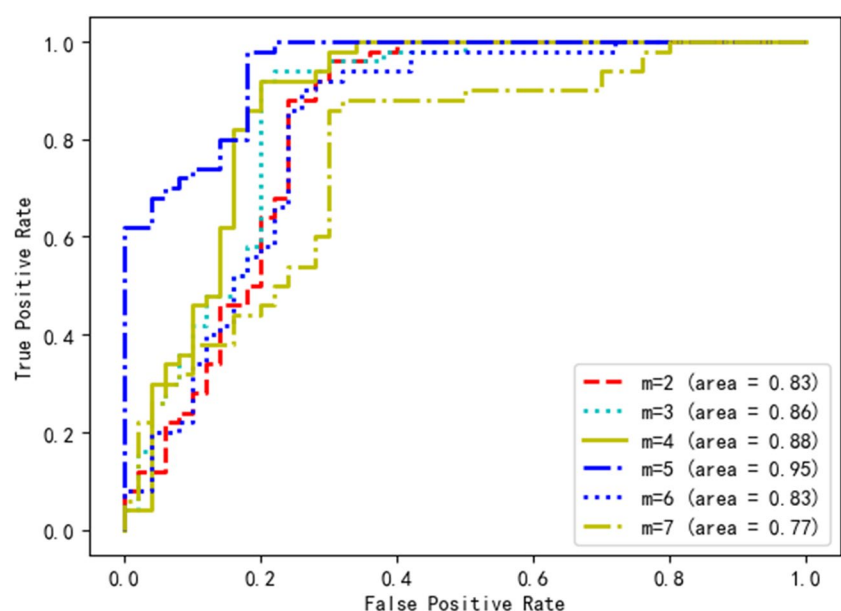
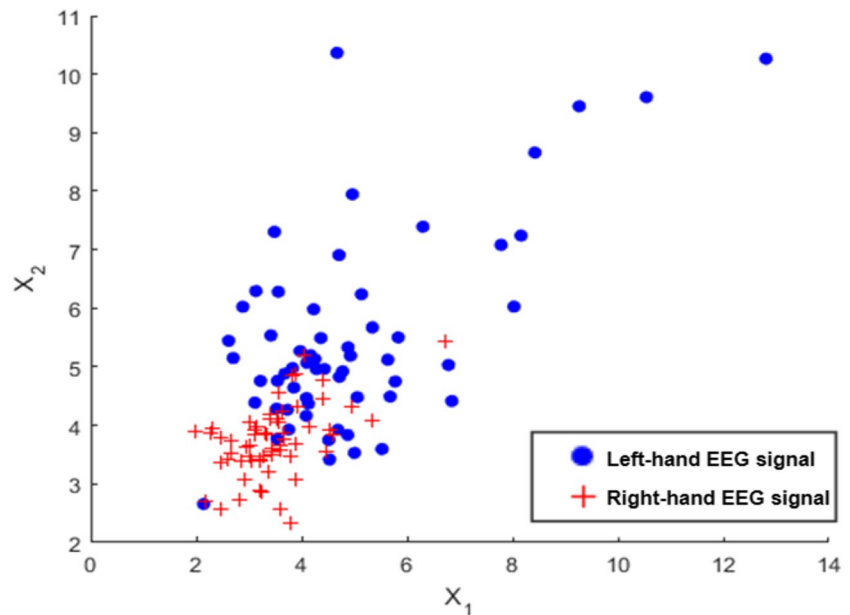


Fig. 4 The 2-dimension features using CSP



3.2 Brain functional network based on PLV

The functional brain network, which outlines the statistical connections between its nodes or brain regions, is a key approach to uncovering the underlying mechanism of information processing in the brain [27]. Various graph measures can be computed to analyze the global and local brain connectivity of networks, providing insight into the structure of the brain network [28]. Global properties, such as global efficiency, clustering coefficient, characteristic path length, etc., represent the communication, segmentation and integration capabilities of the human brain. Meanwhile, local properties, such as betweenness centrality and degree centrality, are only applicable to the specific node in question.

By utilizing Eq. (3) to calculate the phase locking values between the nodes of the left-hand and right-hand EEG signals, the mean adjacency matrix and binarization processing can be obtained by adding the mean values of the adjacency matrices of the signals within the same class, as illustrated in Fig. 5.

Figure 5a and c illustrates the phase synchronization between the channels of the brain functional network obtained from the EEG signals collected from the subjects using the left and right hands, respectively. This allows for the separation of the brain functional network of the left- and right-hand motor imagery data based on the phase locking value. Research has demonstrated that the global differences of different brain functional networks are smaller than the local differences for EEG data [29]. In this study, we have chosen two global features and one local feature of complex network analysis to be analyzed.

The average values of node degree, clustering coefficient and betweenness centrality feature vectors of different classes of 10 subjects were obtained respectively. Subsequently, the

topological attributes of the brain network based on PLV were obtained, and the feature data in the multidimensional space was projected onto the two-dimensional plane by PCA. The distribution of the two-dimensional features is illustrated in Fig. 6. Generally speaking, the features of left- and right-hand EEG signals extracted by PLV-based brain functional network have a certain degree of separability, however, the two-dimensional visualization performance is not satisfactory.

3.3 EEG feature fusion based on UMAP algorithm

3.3.1 Classified performance

In order to ascertain the optimal embedding dimension of UMAP for motor imagery EEG data, while minimizing computational effort, this study randomly selected 5 subjects and applied the UMAP algorithm to reduce their features in various dimensions. Subsequently, a k-nearest neighbors (KNN) classifier was utilized to classify the results after dimension reduction, and the tenfold cross-validation method was employed, which randomly partitions a dataset into ten parts: nine parts are used for training and one part is used for testing. The average of the classification accuracy of the results of ten tests is used to provide an estimation of the accuracy of the algorithm. Each test has its own classification accuracy, and the average of these results is used to give an indication of the overall accuracy of the algorithm. Table 1 presents the classification accuracy of the UMAP algorithm when applied to dimensionality reduction with varying embedded dimensions. The results demonstrate the accuracy of the algorithm when applied to different embedded dimensions, where the values in bold indicate that the

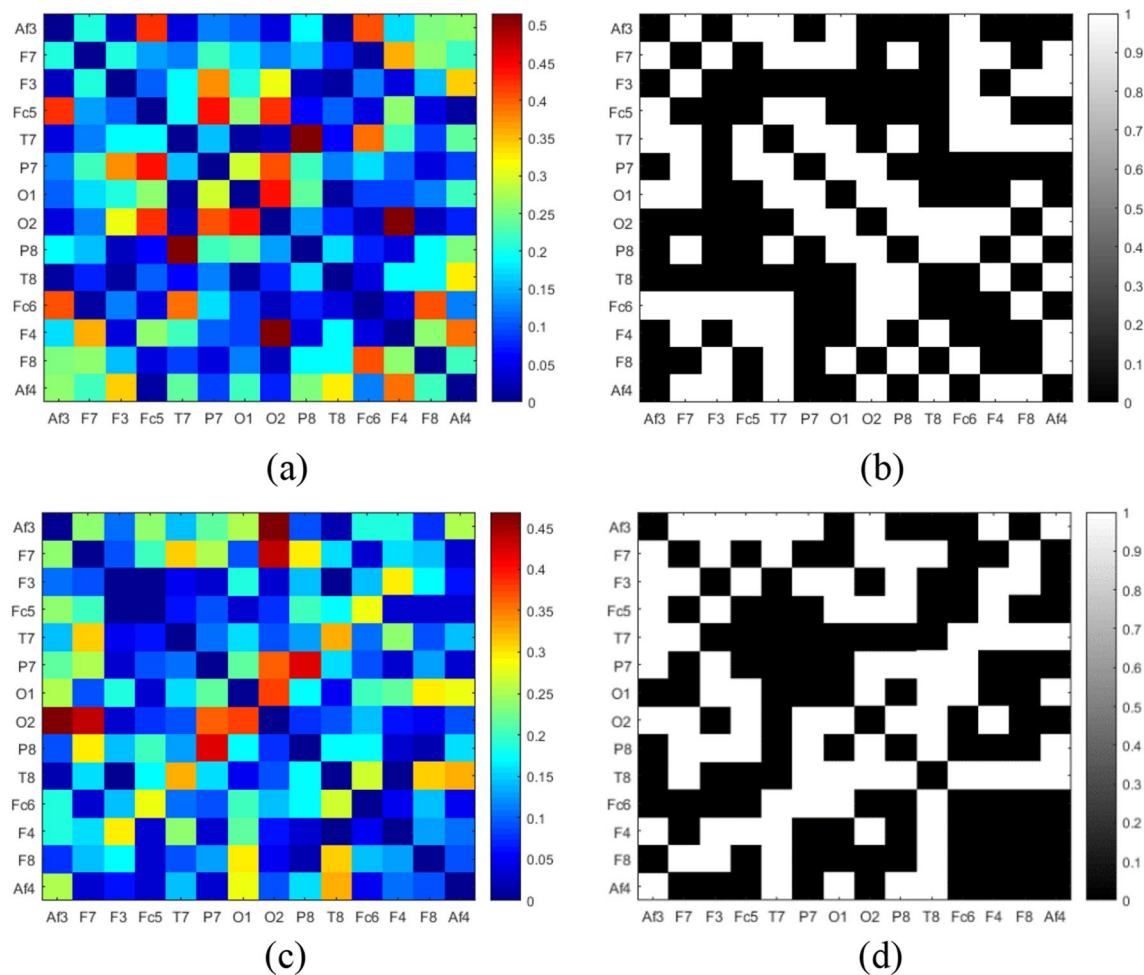


Fig. 5 Adjacency matrix of brain networks based on PLV. **a** Left-hand adjacency matrix patch map. **b** Left-hand binary adjacency matrix patch map. **c** Right-hand adjacency matrix patch map. **d** Right-hand binary adjacency matrix patch map

subject achieved the optimal classification accuracy under the current conditions.

According to Table 1, subjects 1, 2, 3 and 4 achieved the highest classification accuracy when the data was reduced to two dimensions, while subject 5 had the highest classification accuracy when the data was reduced to three dimensions. Consequently, it is recommended to use the UMAP algorithm to reduce the data dimension to two and three dimensions. In this study, the UMAP approach was chosen to reduce the feature vector U_i of EEG data from 52 to 2 dimensions, in order to reduce the computational complexity and time for the subsequent recognition process [30].

To further investigate the necessity and feasibility of feature fusion, five features were extracted from all 10 subjects and classified by the KNN classifier. The recognition results before and after feature fusion for 10 subjects are presented in Table 2 and Table 3, respectively.

The experimental results indicate that the multi-feature fusion algorithm achieved the highest ACC value for

subjects S1, S2, S3, S4, S5, S9 and S10. For subjects S6 and S8, the CSP algorithm yielded the highest ACC value, and for subject S7, the betweenness centrality feature recognition had the highest ACC value. In conclusion, the mean ACC and AUC values obtained from the multi-feature fusion algorithm were 0.9279 and 0.9610, respectively, which were higher than the results obtained when CSP features, node degree features, clustering coefficient features and betweenness centrality features were extracted separately. As a result, the recognition performance after feature dimensionality reduction by the UMAP algorithm is significantly improved compared to the pre-fusion, with the average classification accuracy rate reaching 96.10% and the highest classification accuracy rate reaching 99.76%.

3.3.2 Visual representation

To further investigate the visualization of feature data after dimensionality reduction using UMAP algorithm [31],

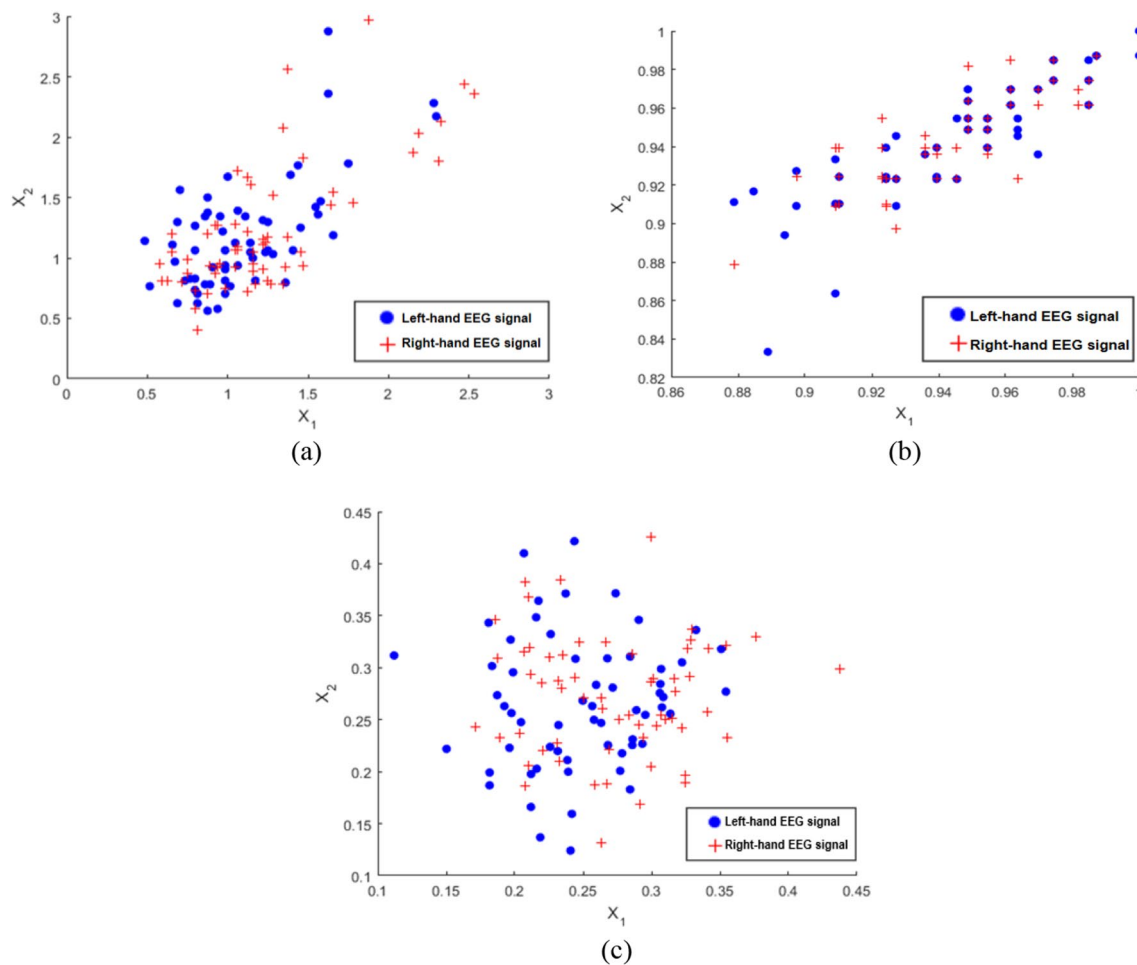


Fig. 6 Spatial distribution features of EEG signals based on PLV network. **a** Node degree spatial distribution features. **b** Clustering coefficient spatial distribution features. **c** Betweenness centrality spatial distribution features

Table 1 Classification accuracy of reducing dimension in different dimensions

Dimension	Subject1	Subject2	Subject3	Subject4	Subject5	AVG
1D	0.7693	0.7574	0.7866	0.7877	0.8016	0.7805
2D	0.9831	0.9615	0.9589	0.9844	0.8770	0.9530
3D	0.9585	0.9106	0.9472	0.9435	0.9460	0.9411
4D	0.9208	0.9043	0.9237	0.9235	0.8939	0.9132

this paper selected the EEG data of one of the subjects and reduced its dimensions to two and three dimensions respectively. The distribution of sample points in two-dimensional plane and three-dimensional space are then shown in Fig. 7, which clearly demonstrates the effectiveness of UMAP in feature dimensionality reduction.

Compared with the spatial distribution of the four features before dimension reduction in previous studies, the left–right hand EEG signals after dimension reduction by UMAP algorithm are significantly different, and the visualization performance is significantly improved. This

indicates that UMAP is an effective method for feature dimensionality reduction and can significantly enhance the recognition performance. It is because the UMAP algorithm makes full use of the nonlinear characteristics of EEG data, shortens the distance between samples of the same class and increases the distance between samples of different classes, thus achieving the purpose of improving the visualization of the data. In general, the detection and visualization performance of the UMAP algorithm after fusion is significantly improved compared with that before fusion, which demonstrates the effectiveness of UMAP in

Table 2 The ACC results before and after feature fusion for 10 subjects

Subject	Accuracy (ACC)				
	Feature fusion	CSP	Node degree	Clustering coefficient	Betweenness centrality
S1	0.9831	0.9072	0.7540	0.7370	0.8730
S2	0.9615	0.8827	0.7500	0.7690	0.8080
S3	0.9589	0.8662	0.8420	0.8420	0.8840
S4	0.9844	0.8746	0.8360	0.7890	0.8590
S5	0.9004	0.8770	0.5980	0.6070	0.6890
S6	0.9068	0.9435	0.8050	0.7880	0.7880
S7	0.9206	0.8783	0.6980	0.7540	0.9320
S8	0.8571	0.9156	0.6640	0.7290	0.8930
S9	0.8788	0.8523	0.6740	0.6590	0.7200
S10	0.9506	0.8949	0.8580	0.8460	0.8700
AVG	0.9279	0.8916	0.7479	0.7520	0.8086

Table 3 The AUC results before and after feature fusion for 10 subjects

Subject	Area under ROC curve (AUC)				
	Feature fusion	CSP	Node degree	Clustering coefficient	Betweenness centrality
S1	0.9948	0.9042	0.8500	0.8300	0.8800
S2	0.9601	0.9303	0.8100	0.8100	0.8700
S3	0.9914	0.9454	0.9000	0.8900	0.9200
S4	0.9976	0.8998	0.8900	0.8800	0.9000
S5	0.9352	0.9153	0.7400	0.6200	0.7900
S6	0.9434	0.9024	0.8800	0.8700	0.8700
S7	0.9403	0.9421	0.8500	0.8400	0.9600
S8	0.9224	0.8699	0.7500	0.8300	0.9300
S9	0.9396	0.8741	0.7900	0.7900	0.7900
S10	0.9851	0.8629	0.9100	0.8700	0.9000
AVG	0.9610	0.9049	0.8370	0.8230	0.8640

feature dimensionality reduction and its ability to significantly enhance the recognition performance.

4 Discussion

In the process of dimensionality reduction in multi-feature fusion, the selection of an appropriate dimensionality reduction technique is also the key to improving the recognition accuracy of motor imagery EEG signals. In this study, three traditional dimensionality reduction algorithms are selected for comparison with the UMAP algorithm, namely PCA, t-SNE (t-distributed stochastic neighbor embedding) and LLE (Locally Linear Embedding). The first traditional

method is PCA, which uses linear transformation to map high-dimensional data to a lower-dimensional space. The second traditional method is t-SNE, which is nonlinear methods that use similarity metrics to transform the data points into lower-dimensional space, and the last method is LLE, which is a nonlinear method to reduce dimension by transforming every data point into a linearly weighted combination of its nearest neighbors. The features of 10 subjects are reduced by different dimensionality reduction algorithms, and the KNN classifier is used for classification. The results show that UMAP outperforms the other three algorithms in terms of recognition accuracy, as shown in Table 4. This indicates that UMAP is an effective method for feature dimensionality reduction and can significantly enhance the recognition performance.

As can be seen from Table 4, the average ACC of the UMAP algorithm is 0.9279, while the average AUC is 0.9610, which is significantly higher than the predictive performance of PCA, t-SNE and LLE. This is likely due to the fact the UMAP algorithm is based on manifold learning and captures the manifold features of EEG signals in the process of dimensionality reduction, while the PCA algorithm, as a linear dimensionality reduction method, achieves lower accuracy due to the loss of data information when projecting data into low dimensions. At the same time, although t-SNE and LLE algorithms are nonlinear dimension reduction methods like UMAP algorithm, the high-dimensional spatial distribution topology of motor imagery EEG signals is not well constructed, and some spatial features are lost, leading to unsatisfactory final performances. Therefore, the UMAP algorithm is more suitable for EEG data than other dimension reduction methods, and its prediction performances are significantly better.

The study also has some limitations. First of all, we did not further verify the feasibility of our method on the public BCI competition motion image data set. Although the data set in our laboratory has certain advantages compared with other data sets, the rationality of the study needs to consider its universal applicability. Therefore, this is the focus of our next research. Secondly, this paper chooses PLV algorithm as the brain network construction algorithm. This is because different from amplitude synchronization method, PLV algorithm can ignore the influence of EEG instantaneous amplitude and is only sensitive to phase. Therefore, it is more suitable for the study of nonlinear and non-stationary EEG signals. However, compared with other algorithms, such as PDC (partial directed coherence) and GC (granger causality), it is unable to detect the directivity between channels. Therefore, these algorithms will also be the focus of future research.

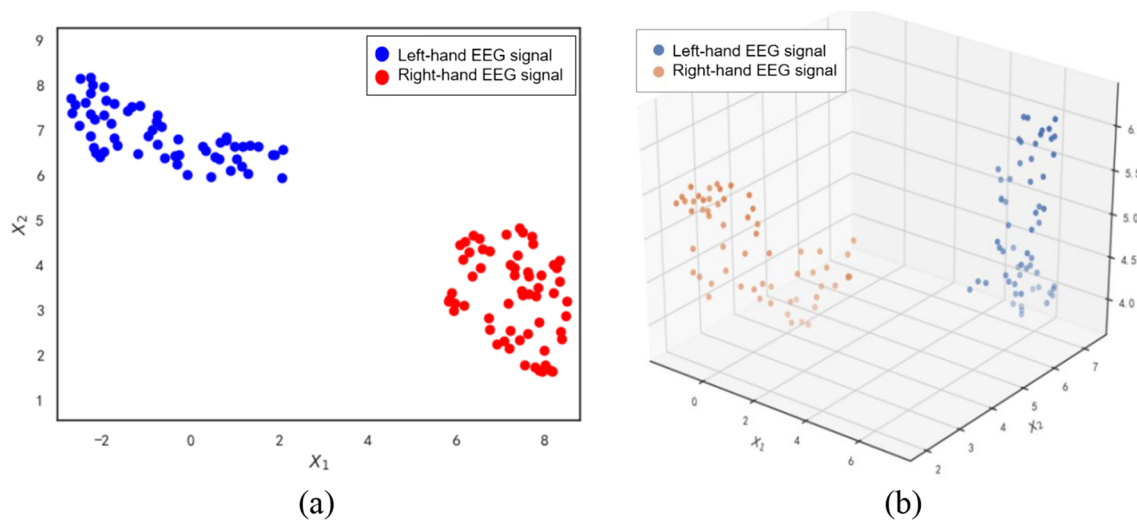


Fig. 7 2D and 3D data visualization for one subject using UMAP algorithm. **a** 2D UMAP **b** 3D UMAP

Table 4 Comparison of prediction results of four methods

Subject	Accuracy (ACC)				Area under ROC curve (AUC)			
	UMAP	PCA	t-SNE	LLE	UMAP	PCA	t-SNE	LLE
S1	0.9831	0.9034	0.8602	0.8600	0.9948	0.8827	0.8453	0.8342
S2	0.9615	0.7602	0.8640	0.8030	0.9601	0.7302	0.8231	0.7652
S3	0.9589	0.7538	0.8823	0.8464	0.9914	0.7122	0.8542	0.8142
S4	0.9844	0.7630	0.8090	0.7752	0.9976	0.7534	0.7932	0.7539
S5	0.8770	0.8416	0.8079	0.8136	0.9352	0.8026	0.7765	0.7865
S6	0.9068	0.8854	0.9284	0.7726	0.9434	0.8712	0.9022	0.7521
S7	0.9206	0.8030	0.9356	0.7822	0.9403	0.7856	0.9143	0.7231
S8	0.8571	0.8948	0.8709	0.8532	0.9224	0.8764	0.8436	0.8486
S9	0.8788	0.8314	0.8261	0.8387	0.9396	0.8012	0.8054	0.8128
S10	0.9506	0.8145	0.8028	0.8291	0.9851	0.7821	0.7839	0.8202
AVG	0.9279	0.8251	0.8587	0.8174	0.9610	0.7997	0.8342	0.7911

5 Conclusion

In this study, we proposed a multi-feature fusion algorithm based on uniform manifold approximation and projection (UMAP) for motor imagery EEG signals. First, the CSP algorithm was used to extract the spatial amplitude information from the EEG signals of each subject. Then, EEG electrodes were considered as nodes of a network, and the PLV matrix of the obtained feature data was calculated as the adjacency matrix of the functional brain network. In terms of feature extraction, three topological attributes of the functional brain network are considered as features. A multi-feature fusion algorithm based on uniform manifold projection is then used to reduce the dimensionality of the features. The features are then classified using a KNN classifier, and the results are evaluated based on accuracy and visualization performance. The results of this study indicate that the fusion

feature has superior separability and higher classification accuracy compared to the single feature. Consequently, the proposed method can improve the classification accuracy and reduce the computational complexity through multi-domain feature extraction and fusion, thus demonstrating its potential for BCI-based applications.

Author contribution Yushan Du and Jiaxin Sui contributed to the conceptualization, systematic review of articles, methodology development, formal analysis, and original writing. Shiwei Wang verified the experimental results and undertook project management. Rongrong Fu participated in the conceptualization, methodology and correction. Chengcheng Jia contributed to the external supervision and correction. All authors read and approved the final manuscript.

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Declarations

Ethics approval All procedures performed in studies involving human participants were in accordance with the ethical standards of the institutional and/or national research committee and with the 1964 Helsinki declaration and its later amendments or comparable ethical standards.

Consent to participate Informed consent was obtained from all individual participants included in the study.

Conflict of interest The authors declare no competing interests.

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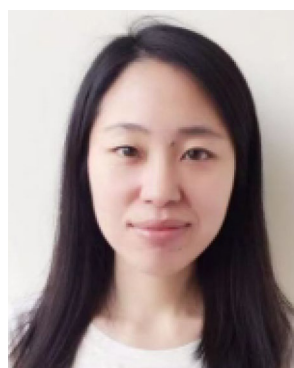
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